EE 533 Information Theory

Week 01

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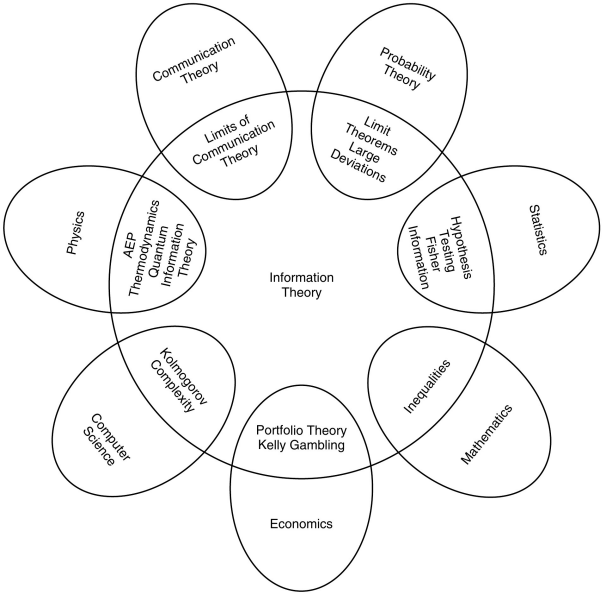


Figure: Relationship of Information Theory to other fields [CT06, fig1.1]

- What is a discrete random variable?
- What is a continuous random variable?
- What is a random variable?
- What is a probability space? (Sample space? Set of events?)

Definition (σ-algebra)

A set $\mathcal{F}$ is a σ-algebra of subsets of Ω iff it satisfies the following three conditions

i) $\Omega \in \mathcal{F}$

ii) $\Omega \setminus \mathcal{E} \in \mathcal{F}$ for all $\mathcal{E} \in \mathcal{F}$.

iii) If $\mathcal{E}_n \in \mathcal{F}$ for all $n \in \mathbb{Z}_+$, then $\cup_{i=1}^{\infty} \mathcal{E}_n \in \mathcal{F}$.

Note that

- $\mathcal{F}$ is closed under countable unions, intersections, and complements.
- If the third condition holds for finite collection of sets rather than countable then $\mathcal{F}$ is an algebra.
- Every σ-algebra is an algebra. But not every algebra is a σ-algebra.
- Third condition is for countable collections not arbitrary collections.

Definition (Measurable Space)

An ordered pair $(\Omega, \mathcal{F})$ is a measurable space iff $\mathcal{F}$ is a σ-algebra of subsets of Ω.

Definition

i) A function $\mu : \mathcal{F} \to [0, \infty]$ is a measure iff μ is countably additive, i.e.

$$\mu\left(\bigcup_{i=1}^{\infty} \mathcal{E}_i\right) = \sum_{i=1}^{\infty} \mu(\mathcal{E}_i)$$

for any countable collection of elements of $\mathcal{F}$ satisfying $\mathcal{E}_i \cap \mathcal{E}_j = \emptyset$ for all $i \neq j$.

ii) A measure μ is a probability measure iff $\mu(\Omega) = 1$.

Definition

$(\Omega, \mathcal{F}, \mathbf{P})$ is a probability space iff $\mathcal{F}$ is σ-algebra of subsets of Ω and $\mathbf{P}$ is a probability measure on $(\Omega, \mathcal{F})$.

Ω	Sample space, certain event
$\omega : \omega \in \Omega$	Outcomes, sample points, elementary events.
$\mathcal{F}$	σ-algebra of events, set of events

Axioms of Probability.

i) $\mathbf{P}(\mathcal{E}) \geq 0$ for all $\mathcal{E} \in \mathcal{F}$.

ii) $\mathbf{P}(\Omega) = 1$

iii) If $\mathcal{E}_i \cap \mathcal{E}_j = \emptyset$ for all $i, j \in \mathbb{Z}_+$ s.t. $i \neq j$ then

$$\mathbf{P}\left(\bigcup_{i=1}^{\infty} \mathcal{E}_i\right) = \sum_{i=1}^{\infty} \mathbf{P}(\mathcal{E}_i)\,.$$

□

Example (Two fair coin tosses)

$$\Omega = \{HH, HT, TH, TT\}$$

$$\mathcal{F} = 2^{\Omega}$$ The power set of Ω, i.e. the set of all subsets of Ω.

$$\mathbf{P}(\{\omega\}) = 1/4$$

Example (Fair coin tosses until an H follows a T or a T follows an H)

$$\Omega = \{HT, TH, HHT, TTH, \cdots\}$$

$$\mathcal{F} = 2^{\Omega}$$

$$\mathbf{P}(\{\omega\}) = 2^{-\ell(\omega)}$$ where $\ell(\omega)$ is the length of the string ω.

Is it possible to use single σ-algebra to describe both discrete and continuous random variables?

Definition

The σ-algebra generated by a collection $\mathcal{S}$ of subsets of a set Ω is the smallest σ-algebra of subsets of Ω that is a superset of $\mathcal{S}$. Such a minimum σ-algebra always exist and it is unique by [Bog07, Proposition 1.2.6.].

Definition

The Borel σ-algebra on $\mathbb{R}$, denoted by $\mathcal{B}(\mathbb{R})$, is the σ-algebra generated by open intervals on the real line.

Since we assign probabilities to all open intervals via $\mathcal{B}(\mathbb{R})$, we know we can describe continuous random variables using $\mathcal{B}(\mathbb{R})$.
On the other hand $\{x\} \in \mathcal{B}(\mathbb{R})$ for all $x \in \mathbb{R}$ because $\{x\}$ is in the intersection of all open intervals containing it that has rational end points.
Thus we can use $\mathcal{B}(\mathbb{R})$ to describe discrete random variables using $\mathcal{B}(\mathbb{R})$.

Definition

On a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ a function $\mathsf{X} : \Omega \to \mathbb{R}$ is a random variable iff $\mathsf{X}^{-1}(\mathcal{A}) \in \mathcal{F}$ for all $\mathcal{A} \in \mathcal{B}(\mathbb{R})$, i.e. X is $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$-measurable.

- In plain English: for all events about X defined by open sets on the real line, I can find an event whose probability I can calculate under $\mathbf{P}$.
  Thus $(\Omega, \mathcal{F}, \mathbf{P})$ and X induce a probability measure $\widetilde{\mathbf{P}}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ via
  $$\widetilde{\mathbf{P}}(\mathcal{A}) := \mathbf{P}\big(\mathsf{X}^{-1}(\mathcal{A})\big) \qquad \forall \mathcal{A} \in \mathcal{B}(\mathbb{R}).$$
- If there exists a countable $\mathcal{X} \subset \mathbb{R}$ s.t. $\mathbf{P}(\mathsf{X} \in \mathcal{X}) = 1$, then X is a discrete random variable.
  $$\mathbf{P}(\mathsf{X} = x) = p_{\mathsf{X}}(x) \qquad \forall x \in \mathcal{X}.$$
- If there exists a non-negative function f_{X} s.t.
  $$\mathbf{P}(\mathsf{X} \leq \tau) = \int_{-\infty}^{\tau} f_{\mathsf{X}}(x)\,\mathrm{d}x \qquad \forall \tau \in \mathbb{R}.$$
  then X is a (absolutely) continuous random variable. We are content with the Riemann integral for this course, but the analysis works for the Lebesgue integral, which is more general.

Information Theory vs/in Probability Theory

- In the probability theory, one is often interested in the random variables and calculations (e.g. algebraic operations) are performed on the random variables themselves.
- In Information theory one often works with the distributions of the random variables and calculations are often performed on the distributions of the random variables themselves.
- Since we are perform calculations on the distributions of the random variables, we need to work with the associated vector space structure.
- For a discrete random variable X, its pmf, i.e. $p_X(X)$, is itself a discrete random variable. (*)
- For a continuous random variable X, its pdf, i.e. $f_X(X)$, in not necessarily a continuous random variable! (Consider a random variable that is uniformly distributed on $[0, 2]$, $f_X(X) = 0.5$ with probability one and $f_X(X)$ is a discrete random variable.)

Vector Spaces

Axiom	Meaning
Associativity of addition	$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
Commutativity of addition	$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
Identity element of addition	There exists an element $\mathbf{0} \in V$, called the *zero vector*, such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in V$.
Inverse elements of addition	For every $\mathbf{v} \in V$, there exists an element $-\mathbf{v} \in V$, called the *additive inverse* of $\mathbf{v}$, such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
Compatibility of scalar multiplication with field multiplication	$a(b\mathbf{v}) = (ab)\mathbf{v}$ [nb 2]
Identity element of scalar multiplication	$1\mathbf{v} = \mathbf{v}$, where 1 denotes the multiplicative identity in F.
Distributivity of scalar multiplication with respect to vector addition	$a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
Distributivity of scalar multiplication with respect to field addition	$(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

Figure: Axioms of Vector Space (wikipedia.org)

Definition (Convex Sets)

A subset $\mathcal{S}$ of a vector space is a *convex set* iff

$$\alpha x + (1-\alpha)y \in \mathcal{S} \qquad \forall x, y \in \mathcal{S}, \alpha \in [0,1].$$

Example

$\left\{x_1^n \in \mathbb{R}^n \text{ s.t. } \sum_{i=1}^n (x_i)^2 \leq 1\right\}$	The unit sphere in $\mathbb{R}^n$
$\left\{x_1^n \in \mathbb{R}^n \text{ s.t. } \sum_{i=1}^n x_i = 1 \text{ and } x_i \geq 0\ \forall i\right\}$	The probability simplex in $\mathbb{R}^n$
$\left\{x_1^\infty \in \mathbb{R}^\infty \text{ s.t. } \sum_{i=1}^\infty x_i = 1 \text{ and } x_i \geq 0\ \forall i\right\}$	The probability simplex in $\mathbb{R}^\infty$
$\left\{f \in \mathcal{L}(\mathbb{R}) \text{ such that } \int f(x)\mathrm{d}x = 1 \text{ and } f(x) \geq 0\right\}$	The set of all pdf on the real line.
$\left\{f \in \mathcal{L}([0,1]) \text{ such that } \left(\int \lvert f(x)\rvert^p \mathrm{d}x\right)^{1/p} \leq 1\right\}$	Convex for $p \geq 1$

Remark

$\mathcal{L}(\mathbb{R})$ stands for the integrable functions on the real line.

Definition (Convex/Concave Functions)

A function $f : \mathcal{S} \to \mathbb{R}$ defined on a convex set $\mathcal{S}$ is *convex* iff

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y) \qquad \forall x, y \in \mathcal{S}, \alpha \in [0,1].$$

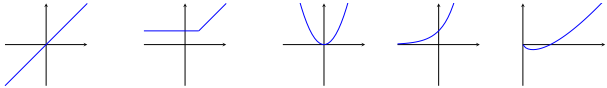
A function $f : \mathcal{S} \to \mathbb{R}$ defined on a convex set $\mathcal{S}$ is *strictly convex* iff

$$f(\alpha x + (1-\alpha)y) < \alpha f(x) + (1-\alpha) f(y) \qquad \forall x, y \in \mathcal{S}, \alpha \in (0,1).$$

A function $f : \mathcal{S} \to \mathbb{R}$ defined on a convex set $\mathcal{S}$ is (*strictly*) *concave* iff $-f$ is (*strictly*) *convex*.

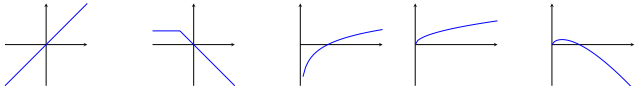
Example

x, $\max\{x, 1\}$, x^2, e^x, $x\log(x)$, are convex functions.



x^2, e^x, $x\log(x)$ are also strictly convex functions.

x, $\min\{-x, 1\}$, $\log(x)$, $\sqrt{x}$, $-x\log(x)$, are concave functions.



$\log(x)$, $\sqrt{x}$, $-x\log(x)$ are also strictly concave functions.

Theorem

Assume $f : \mathbb{R} \to \mathbb{R}$ is a twice continuously differentiable function.

a) *If $f'' \geq 0$ on a convex $I \subset \mathbb{R}$ then f is convex on I.*

b) *If $f'' > 0$ on a convex $I \subset \mathbb{R}$ then f is strictly convex on I.*

Proof.

By Taylor's series expansion, for any x, y, there exists z between x and y s.t.

$$f(y) = f(x) + f'(x)(y - x) + \frac{f''(z)}{2}(y - x)^2$$

Let $x_\alpha = \alpha x_1 + (1 - \alpha)x_0$ for any $x_1, x_0 \in I$ and $\alpha \in [0, 1]$. Then

$$f(x_1) = f(x_\alpha) + f'(x_\alpha)(x_1 - x_\alpha) + \frac{f''(z_1)}{2}(x_1 - x_\alpha)^2$$

$$f(x_0) = f(x_\alpha) + f'(x_\alpha)(x_0 - x_\alpha) + \frac{f''(z_0)}{2}(x_0 - x_\alpha)^2$$

$$\alpha f(x_1) + (1 - \alpha)f(x_0) = f(x_\alpha) + \alpha(1 - \alpha)\frac{(1-\alpha)f''(z_1) + \alpha f''(z_0)}{2}(x_1 - x_0)^2$$

$\square$

Theorem

Assume $f : \mathbb{R} \to \mathbb{R}$ is a twice continuously differentiable function.

a) *If $f'' \leq 0$ on a concave $I \subset \mathbb{R}$ then f is concave on I.*

b) *If $f'' < 0$ on a concave $I \subset \mathbb{R}$ then f is strictly concave on I.*

Proof.

f is (strictly) concave iff $-f$ is (strictly) convex. □

Theorem

a) *If $f_i : \mathcal{S} \to \mathbb{R}$ are convex and $\alpha_i \geq 0$, then $\sum_{i=1}^{k} \alpha_i f_i$ is a convex function.*

b) *If $f_i : \mathcal{S} \to \mathbb{R}$ are convex, then $f(s) := \max_i f_i(s)$ is a convex function.*

c) *If $g : \mathcal{A} \to \mathcal{S}$ is a linear (or an affine) function and $f : \mathcal{S} \to \mathbb{R}$ is a convex function then $f \circ g : \mathcal{A} \to \mathbb{R}$ is a convex function.*

Proof.

All assertions of the theorem be confirmed by substitution. □

Theorem (Jensen's Inequality)

For any $\mathcal{S}$ valued random object (variable) X and convex function $f : \mathcal{S} \to \mathbb{R}$:

$$f(\mathbf{E}[\mathsf{X}]) \leq \mathbf{E}[f(\mathsf{X})].$$

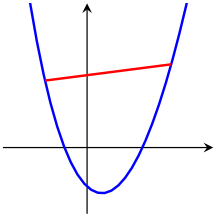
If f is strictly convex then inequality is strict unless $\mathsf{X} = \mathbf{E}[\mathsf{X}]$ a.s.

Proof for the discrete case:

If X takes only two values theorem follows from the definition.

If theorem holds for the case when X takes only $k-1$ distinct values, then for a random variable that takes k distinct values we have

$$\begin{aligned}
\mathbf{E}[f(\mathsf{X})] &= p_{\mathsf{X}}(x_k)\, f(x_k) + (1 - p_{\mathsf{X}}(x_k)) \sum_{i=1}^{k-1} \frac{p_{\mathsf{X}}(x_i)}{1-p_{\mathsf{X}}(x)_k} f(x_i) \\
&\geq p_{\mathsf{X}}(x_k)\, f(x_k) + (1 - p_{\mathsf{X}}(x_k)) f\left(\sum_{i=1}^{k-1} \frac{p_{\mathsf{X}}(x_i)}{1-p_{\mathsf{X}}(x_k)} x_i\right) \\
&\geq f\left(\sum_{i=1}^{k} p_{\mathsf{X}}(x_i)\, x_i\right) \\
&= f(\mathbf{E}[\mathsf{X}])
\end{aligned}$$



Then theorem holds for discrete random variable by induction. □

For a more general proof see [Dud02, p. 348].

Definition

The entropy $H(\mathsf{X})$ of a discrete random variable X is defined as

$$H(\mathsf{X}) := \sum_x p_{\mathsf{X}}(x) \log \frac{1}{p_{\mathsf{X}}(x)} \qquad = \mathbf{E}\left[\log \frac{1}{p_{\mathsf{X}}(\mathsf{X})}\right]$$

<u>Convention:</u> $0 \log \frac{1}{0} = 0$. Note that $\lim_{\tau \downarrow 0} \tau \log \frac{1}{\tau} = 0$.
Basis of the logarithm (always greater than one) determines the units:

$$\log_2 \leftrightarrow \text{bits}$$
$$\log_e \ (i.e.\ \ln) \leftrightarrow \text{nats}$$
$$\log_{256} \leftrightarrow \text{bytes}$$

Theorem (Non-negativity of the entropy)

$H(\mathsf{X}) \geq 0$. $H(\mathsf{X}) = 0$ *iff* X *is constant, i.e.* $p_{\mathsf{X}}(x_0) = 1$ *for some* x_0.

Theorem (Invariance under relabeling)

$H(\mathsf{X}) = H(f(\mathsf{X}))$ *for any one-to-one function* f.

Thus definition applies to any random object that takes its value from a countable set.

Example

$X \in \{0, 1\}$, $\mathbf{P}(X = 0) = p$, and $\mathbf{P}(X = 1) = 1 - p$

$$H(X) = \hbar(p) := p \log \frac{1}{p} + (1 - p) \log \frac{1}{1-p}$$

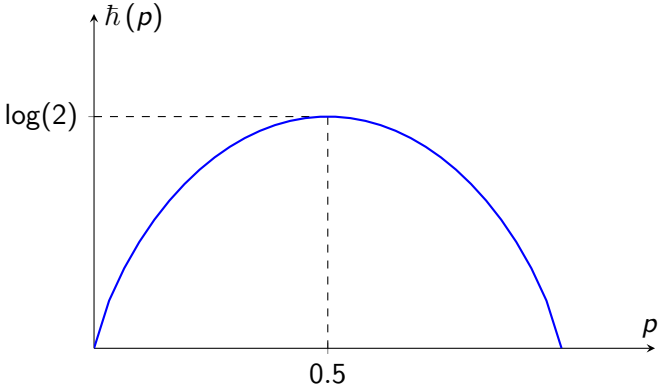


Figure: Binary Entropy Function $\hbar(p)$

Example

$$\mathsf{X} \in \{a, b, c, d\} \text{ and } p_{\mathsf{X}}(x) = \begin{cases} 1/2 & x = a \\ 1/4 & x = b \\ 1/8 & x = c \\ 1/8 & x = d \end{cases}$$

$$H(\mathsf{X}) = 1/2 \log 2 + 1/4 \log 4 + 1/8 \log 8 + 1/8 \log 8 \qquad = \frac{7}{4} \log 2$$

How many yes-no questions, on average, do I need to ask to learn X?
Let us consider a particular strategy "$\mathsf{X} = a$? Is $\mathsf{X} = b$? Is $\mathsf{X} = c$?"

$$\mathbf{E}[\mathsf{N}] = (1/2) \times 1 + (1/4) \times 2 + (1/8) \times 3 + (1/8) \times 3 \qquad = 7/4$$

$$H(\mathsf{X}) \leq \mathbf{E}[\mathsf{N}] \leq H(\mathsf{X}) + 1.$$

Is $H(\mathrm{X})$ infinite, whenever there are countably infinite possible values for X?

Example

$\mathrm{X} \in \mathbb{Z}_{\geq 0}$ and $\mathbf{P}(\mathrm{X} = k) = p(1-p)^k$.

$$\begin{aligned}
H(\mathrm{X}) &= \sum_{k=0}^{\infty} p(1-p)^k \log \tfrac{1}{p(1-p)^k} \\
&= \log \tfrac{1}{p} + p(1-p)\left(\log \tfrac{1}{1-p}\right) \sum_{k=0}^{\infty} k(1-p)^{k-1} \\
&= \log \tfrac{1}{p} + p(1-p)\left(\log \tfrac{1}{1-p}\right) \left(-\tfrac{\mathrm{d}}{\mathrm{d}\tau} \sum_{k=0}^{\infty} (1-\tau)^k \Big|_{\tau=p}\right) \\
&= \log \tfrac{1}{p} + p(1-p)\left(\log \tfrac{1}{1-p}\right) \left(-\tfrac{\mathrm{d}}{\mathrm{d}\tau} \tfrac{1}{1-(1-\tau)} \Big|_{\tau=p}\right) \\
&= \log \tfrac{1}{p} + \tfrac{1-p}{p} \log \tfrac{1}{1-p} \\
&= \hbar(p)/p
\end{aligned}$$

Example

$\mathrm{X} \in \{2, 3, \ldots\}$ and $\mathbf{P}(\mathrm{X} = k) = \frac{c}{k \log^2 k}$ where $c := \left(\sum_{k=2}^{\infty} \frac{1}{k \log^2 k}\right)^{-1}$.

$$H(\mathrm{X}) = \sum_{k=2}^{\infty} \frac{c}{k \log^2 k} \log \frac{k \log^2 k}{c} \quad \geq \sum_{k > 2^{\sqrt{c}}}^{\infty} \frac{c}{k \log^2 k} \log \frac{k \log^2 k}{c} \quad \geq \sum_{k > 2^c}^{\infty} \frac{c}{k \log k} = \infty$$

Theorem

a) $H(\mathsf{X})$ *is a strictly concave function of* p_{X}.

b) $H(\mathsf{X}) \leq \log |\mathrm{supp}(p_{\mathsf{X}})|$ *and the equality is strict unless* p_{X} *is uniform.*

Proof.

a) $H(\mathsf{X}) = \sum_x f(p_{\mathsf{X}}(x))$ where $f(z) := z \ln \frac{1}{z}$.
Since $p_{\mathsf{X}} \rightsquigarrow p_{\mathsf{X}}(x)$ is linear, $H(\mathsf{X})$ is strictly concave in p_{X} provided that f is strictly concave.
$f'(z) := -1 + \ln \frac{1}{z}$
$f''(z) := -\frac{1}{z} \Rightarrow f$ is strictly concave.

b) Let $\mathsf{Y} := \frac{1}{p_{\mathsf{X}}(\mathsf{X})}$ where $\mathrm{supp}(p_{\mathsf{X}}) := \{x : p_{\mathsf{X}}(x) > 0\}$. Then

$$H(\mathsf{X}) = \mathbf{E}[\log \mathsf{Y}] \qquad \mathbf{E}[\mathsf{Y}] = |\mathrm{supp}(p_{\mathsf{X}})|$$

Strict concavity of logarithm function and Jensen's inequality imply

$$\mathbf{E}[\log(\mathsf{Y})] \leq \log(\mathbf{E}[\mathsf{Y}]).$$

HW: Alternative proof considering the permutations of the pmf and the strict concavity of $H(\mathsf{X})$ in p_{X}. □

Definition (The Joint Entropy and The Conditional Entropy)

The joint entropy $H(\mathrm{X},\mathrm{Y})$ of discrete random variables X and Y is defined as

$$H(\mathrm{X},\mathrm{Y}) := \sum_{x,y} p_{\mathrm{X},\mathrm{Y}}(x,y) \log \frac{1}{p_{\mathrm{X},\mathrm{Y}}(x,y)} \qquad = \mathbf{E}\left[\log \frac{1}{p_{\mathrm{X},\mathrm{Y}}(\mathrm{X},\mathrm{Y})}\right].$$

The conditional entropy $H(\mathrm{X}|\mathrm{Y})$ of X given Y is defined as

$$H(\mathrm{X}|\mathrm{Y}) := \sum_{x,y} p_{\mathrm{X},\mathrm{Y}}(x,y) \log \frac{1}{p_{\mathrm{X}|\mathrm{Y}}(x|y)} \qquad = \mathbf{E}\left[\log \frac{1}{p_{\mathrm{X}|\mathrm{Y}}(\mathrm{X}|\mathrm{Y})}\right].$$

Recall $\mathbf{E}[\mathrm{X}|\mathrm{Y}]$ is a random variable, $H(\mathrm{X}|\mathrm{Y})$ is not!

$$H(\mathrm{X}|\mathrm{Y}=y) := \sum_{x} p_{\mathrm{X}|\mathrm{Y}}(x|y) \log \frac{1}{p_{\mathrm{X}|\mathrm{Y}}(x|y)} \qquad = \mathbf{E}\left[\log \frac{1}{p_{\mathrm{X}|\mathrm{Y}}(\mathrm{X}|\mathrm{Y})}\middle|\mathrm{Y}=y\right].$$

Instead of $\mathbf{E}[\mathrm{X}] = \mathbf{E}[\mathbf{E}[\mathrm{X}|\mathrm{Y}]]$ we have:

$$H(\mathrm{X}|\mathrm{Y}) = \sum_{y} p_{\mathrm{Y}}(y)\, H(\mathrm{X}|\mathrm{Y}=y) \qquad = \mathbf{E}_{y\sim p_{\mathrm{Y}}}\left[H(\mathrm{X}|\mathrm{Y}=y)\right].$$

Theorem (Chain Rule For The Entropy)

$$H(X,Y) = H(X) + H(Y|X) \qquad = H(Y) + H(X|Y)$$

Proof.

$$\begin{aligned} H(X,Y) &= \mathbf{E}\left[\log \frac{1}{p_{X,Y}(X,Y)}\right] \\ &= \mathbf{E}\left[\log \frac{1}{p_X(X)p_{Y|X}(Y|X)}\right] \\ &= \mathbf{E}\left[\log \frac{1}{p_X(X)}\right] + \mathbf{E}\left[\log \frac{1}{p_{Y|X}(Y|X)}\right] \\ &= H(X) + H(Y|X) \end{aligned}$$

□

Corollary: $H(X) \geq H(f(X))$

Why?When does the equality hold?

Question: $H(Y|X) \stackrel{?}{=} H(X|Y)$

Theorem (Conditioning Reduces The Entropy)

$$H(\mathsf{X}|\mathsf{Y}) \leq H(\mathsf{X})$$

The inequality is strict unless X *and* Y *are independent.*

Proof.

$$\left.\begin{aligned} H(\mathsf{X}|\mathsf{Y}=y) &:= \textstyle\sum_x p_{\mathsf{X}|\mathsf{Y}}(x|y) \log \frac{1}{p_{\mathsf{X}|\mathsf{Y}}(x|y)} \\ H(P_{\mathsf{X}}) &:= \textstyle\sum_x p_{\mathsf{X}}(x) \log \frac{1}{p_{\mathsf{X}}(x)} \end{aligned}\right\} \Rightarrow H(\mathsf{X}|\mathsf{Y}=y) = H\big(p_{\mathsf{X}|\mathsf{Y}}(\cdot|y)\big)$$

Thus $H(\mathsf{X}|\mathsf{Y}) = \sum_y p_{\mathsf{Y}}(y)\, H(\mathsf{X}|\mathsf{Y}=y)$ implies

$$\begin{aligned} H(\mathsf{X}|\mathsf{Y}) &= \sum\nolimits_y p_{\mathsf{Y}}(y)\, H\big(p_{\mathsf{X}|\mathsf{Y}}(\cdot|y)\big) \\ &\leq H\Big(\sum\nolimits_y p_{\mathsf{Y}}(y)\, p_{\mathsf{X}|\mathsf{Y}}(\cdot|y)\Big) && \text{by Jensen's ineq. and concavity of ent.} \\ &= H(p_{\mathsf{X}}) \\ &= H(\mathsf{X})\,. \end{aligned}$$

□

- Jensen's inequality: if f is a convex function, then $f(\mathbf{E}[\mathsf{X}]) \leq \mathbf{E}[f(\mathsf{X})]$.
- f is a concave function iff $-f$ is a convex function.

Definition (The Mutual Information)

The Mutual Information between discrete random variables X and Y is

$$
\begin{aligned}
I(\mathrm{X};\mathrm{Y}) &:= \sum_{x,y} p_{\mathrm{X},\mathrm{Y}}(x,y) \log \frac{p_{\mathrm{X},\mathrm{Y}}(x,y)}{p_{\mathrm{X}}(x)p_{\mathrm{Y}}(y)} && = \mathbf{E}\left[\log \frac{p_{\mathrm{X},\mathrm{Y}}(\mathrm{X},\mathrm{Y})}{p_{\mathrm{X}}(\mathrm{X})p_{\mathrm{Y}}(\mathrm{Y})}\right] \\
&= \sum_{x,y} p_{\mathrm{X},\mathrm{Y}}(x,y) \log \frac{p_{\mathrm{X}|\mathrm{Y}}(x|y)}{p_{\mathrm{X}}(x)} && = \mathbf{E}\left[\log \frac{p_{\mathrm{X}|\mathrm{Y}}(\mathrm{X}|\mathrm{Y})}{p_{\mathrm{X}}(\mathrm{X})}\right] && (1) \\
&= H(\mathrm{X}) - H(\mathrm{X}|\mathrm{Y}) && (2) \\
&= H(\mathrm{X}) + H(\mathrm{Y}) - H(\mathrm{X},\mathrm{Y}) \\
&= H(\mathrm{Y}) - H(\mathrm{Y}|\mathrm{X}) && (3)
\end{aligned}
$$

Theorem

a) $I(\mathrm{X};\mathrm{Y}) = I(\mathrm{Y};\mathrm{X})$ *(definition of $I(\mathrm{X};\mathrm{Y})$)*

b) $I(\mathrm{X};\mathrm{X}) = H(\mathrm{X})$ *($H(\mathrm{X}|\mathrm{X}) = 0$)*

c) $I(\mathrm{X};\mathrm{Y}) \geq 0$, *with equality iff* X *and* Y *are independent, i.e.* $p_{\mathrm{X},\mathrm{Y}} = p_{\mathrm{X}}p_{\mathrm{Y}}$. *(conditioning reduces the entropy)*

d) $I(\mathrm{X};\mathrm{Y}) \leq H(\mathrm{X}) \wedge H(\mathrm{Y})$ *with equality iff either* $H(\mathrm{X}|\mathrm{Y}) = 0$ *or* $H(\mathrm{Y}|\mathrm{X}) = 0$. *((2),$H(\mathrm{X}|\mathrm{Y}) \geq 0$,(3),$H(\mathrm{Y}|\mathrm{X}) \geq 0$)*

Definition (The Conditional Mutual Information)

$$
\begin{aligned}
I(\mathrm{X};\mathrm{Z}|\mathrm{Y}) &:= \sum_{x,y,z} p_{\mathrm{X},\mathrm{Y},\mathrm{Z}}(x,y,z)\log\frac{p_{\mathrm{X},\mathrm{Z}|\mathrm{Y}}(x,z|y)}{p_{\mathrm{X}|\mathrm{Y}}(x|y)p_{\mathrm{Z}|\mathrm{Y}}(z|y)} &&= \mathbf{E}\left[\log\frac{p_{\mathrm{X},\mathrm{Z}|\mathrm{Y}}(\mathrm{X},\mathrm{Z}|\mathrm{Y})}{p_{\mathrm{X}|\mathrm{Y}}(\mathrm{X}|\mathrm{Y})p_{\mathrm{Z}|\mathrm{Y}}(\mathrm{Z}|\mathrm{Y})}\right]\\
&= \sum_{x,y,z} p_{\mathrm{X},\mathrm{Y},\mathrm{Z}}(x,y,z)\log\frac{p_{\mathrm{X}|\mathrm{Y},\mathrm{Z}}(x|y,z)}{p_{\mathrm{X}|\mathrm{Y}}(x|y)} &&= \mathbf{E}\left[\log\frac{p_{\mathrm{X}|\mathrm{Y},\mathrm{Z}}(\mathrm{X}|\mathrm{Y},\mathrm{Z})}{p_{\mathrm{X}|\mathrm{Y}}(\mathrm{X}|\mathrm{Y})}\right]\\
&= H(\mathrm{X}|\mathrm{Y}) + H(\mathrm{Z}|\mathrm{Y}) - H(\mathrm{X},\mathrm{Z}|\mathrm{Y})\\
&= H(\mathrm{X}|\mathrm{Y}) - H(\mathrm{X}|\mathrm{Y},\mathrm{Z})\\
&= H(\mathrm{Z}|\mathrm{Y}) - H(\mathrm{Z}|\mathrm{Y},\mathrm{X})\\
I(\mathrm{X};\mathrm{Z}|\mathrm{Y}=y) &:= \sum_{x,z} p_{\mathrm{X},\mathrm{Z}|\mathrm{Y}}(x,z|y)\log\frac{p_{\mathrm{X},\mathrm{Z}|\mathrm{Y}}(x,z|y)}{p_{\mathrm{X}|\mathrm{Y}}(x|y)p_{\mathrm{Z}|\mathrm{Y}}(z|y)} &&= \mathbf{E}\left[\log\frac{p_{\mathrm{X},\mathrm{Z}|\mathrm{Y}}(\mathrm{X},\mathrm{Z}|y)}{p_{\mathrm{X}|\mathrm{Y}}(\mathrm{X}|y)p_{\mathrm{Z}|\mathrm{Y}}(\mathrm{Z}|y)}\middle|\mathrm{Y}=y\right]\\
&= H(\mathrm{X}|\mathrm{Y}=y) + H(\mathrm{Z}|\mathrm{Y}=y) - H(\mathrm{X},\mathrm{Z}|\mathrm{Y}=y)\\
I(\mathrm{X};\mathrm{Z}|\mathrm{Y}) &= \mathbf{E}_{y\sim p_{\mathrm{Y}}}\left[I(\mathrm{X};\mathrm{Z}|\mathrm{Y}=y)\right]
\end{aligned}
$$

Theorem

$I(\mathrm{X};\mathrm{Z}|\mathrm{Y}) \geq 0$, with equality iff X and Z are conditionally independent given Y, i.e. $p_{\mathrm{X},\mathrm{Z}|\mathrm{Y}} = p_{\mathrm{X}|\mathrm{Y}}p_{\mathrm{Z}|\mathrm{Y}}$.

Proof.

Use the corresponding thm. for the mutual information on $I(\mathrm{X};\mathrm{Z}|\mathrm{Y}=y)$. □

Theorem (Chain Rule For The Mutual Information)

$$\begin{aligned} I(X,Z;Y) &= I(Z;Y) + I(X;Y|Z) \\ &= I(X;Y) + I(Z;Y|X) \end{aligned}$$

Proof.

$$\begin{aligned} I(X,Z;Y) &= H(Y) - H(Y|X,Z) \\ &= H(Y) - H(Y|Z) + H(Y|Z) - H(Y|X,Z) \\ &= I(Z;Y) + I(X;Y|Z) \end{aligned}$$

□

Definition (Joint/Conditional Entropy)

Given a discrete random vector $\mathsf{X}_1^k = (\mathsf{X}_1, \ldots, \mathsf{X}_k)$,

$$H\left(\mathsf{X}_1^k\right) := -\sum_{x_1^k} p_{\mathsf{X}_1^k}\left(x_1^k\right) \log p_{\mathsf{X}_1^k}\left(x_1^k\right) \qquad = -\mathbf{E}\left[\log p_{\mathsf{X}_1^k}\left(\mathsf{X}_1^k\right)\right]$$

$$H\left(\mathsf{X}_{i+1}^k \middle| \mathsf{X}_1^i\right) := -\sum_{x_1^k} p_{\mathsf{X}_1^k}\left(x_1^k\right) \log p_{\mathsf{X}_{i+1}^k|\mathsf{X}_1^i}\left(x_{i+1}^k | x_1^i\right) \qquad = -\mathbf{E}\left[\log p_{\mathsf{X}_{i+1}^k|\mathsf{X}_1^i}\left(\mathsf{X}_{i+1}^k | \mathsf{X}_1^i\right)\right]$$

Theorem (Chain Rules)

$$H\left(\mathsf{X}_1^k\right) = H(\mathsf{X}_1) + \sum_{i=1}^{k-1} H\left(\mathsf{X}_{i+1} \middle| \mathsf{X}_1^i\right)$$

$$I\left(\mathsf{X}_1^k; \mathsf{Y}\right) = I(\mathsf{X}_1; \mathsf{Y}) + \sum_{i=1}^{k-1} I\left(\mathsf{X}_{i+1}; \mathsf{Y} \middle| \mathsf{X}_1^i\right)$$

HW: Prove the theorem in two ways: (a) by induction and (b) by writing joint probability mass functions in terms of the conditionals and a marginal.

Corollary (Independence Bound)

$$H\left(\mathsf{X}_1^k\right) \leq \sum_{i=1}^{k} H(\mathsf{X}_i)$$

HW: When does $I\left(\mathsf{X}_1^k; \mathsf{Y}\right) \leq \sum_{i=1}^{k} I(\mathsf{X}_i; \mathsf{Y})$ hold? Is it a good bound?

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